

Data Analytics Project 2

[Exploratory Data Analysis Report]

Introduction

This report presents the findings of an Exploratory Data Analysis (EDA) conducted on a cleaned e-commerce sales dataset. The analysis was performed in Python using pandas, NumPy, Matplotlib and Seaborn and covers four major analytical dimensions: descriptive statistics, distribution analysis, trend and pattern analysis, and correlation analysis.

The dataset used is the output of Project 1 (data cleaning) and contains transaction-level records including product, pricing, date, order status, and referral source information.

Descriptive Statistics

Summary statistics were computed for all numeric columns using `df.describe()`. A custom table was additionally printed showing the mean, median and skewness for each column, with a '(skewed)' flag applied when the absolute skewness exceeded 1.0.

Key Metrics Captured

- Count, mean, standard deviation, minimum and maximum values.
- 25th, 50th (median) and 75th percentile values.
- Skewness coefficient - values above $|1|$ indicate a significantly skewed distribution.

Skewness analysis is particularly important for revenue-related columns like `TotalPrice`, where extreme transactions (very large or very small orders) can pull the mean away from the median, signalling a non-normal distribution.

Descriptive Statistics for Numerical Columns				
	Quantity	UnitPrice	ItemsInCart	TotalPrice
count	1200.00	1200.00	1200.00	1200.00
mean	2.95	356.41	5.48	1053.97
std	1.41	197.18	2.28	819.86
min	1.00	11.39	1.00	11.39
25%	2.00	186.06	4.00	410.52
50%	3.00	364.21	5.00	823.62
75%	4.00	521.57	7.00	1578.48
max	5.00	699.93	10.00	3456.40

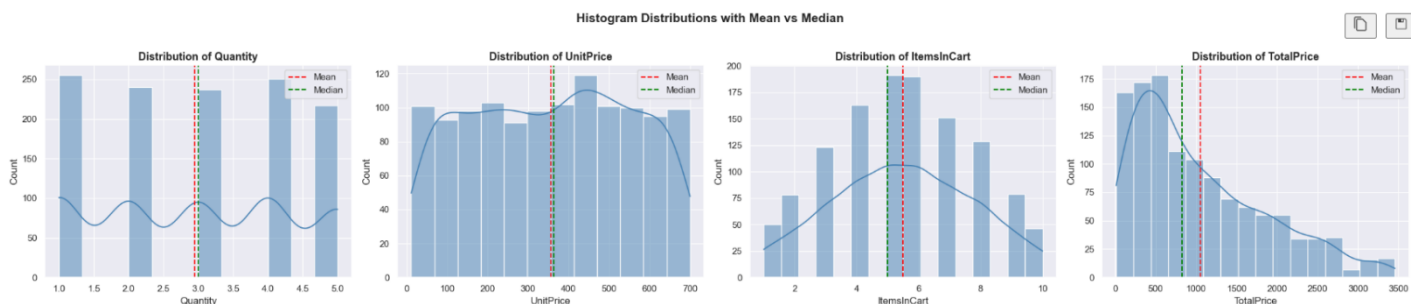
Column	Mean	Median	Skew
Quantity	2.95	3.00	0.03
UnitPrice	356.41	364.21	-0.03
ItemsInCart	5.49	5.00	0.00
TotalPrice	1053.97	823.62	0.89

Distribution Analysis

Distribution plots were generated for all numerical columns to visually inspect the shape and spread of the data.

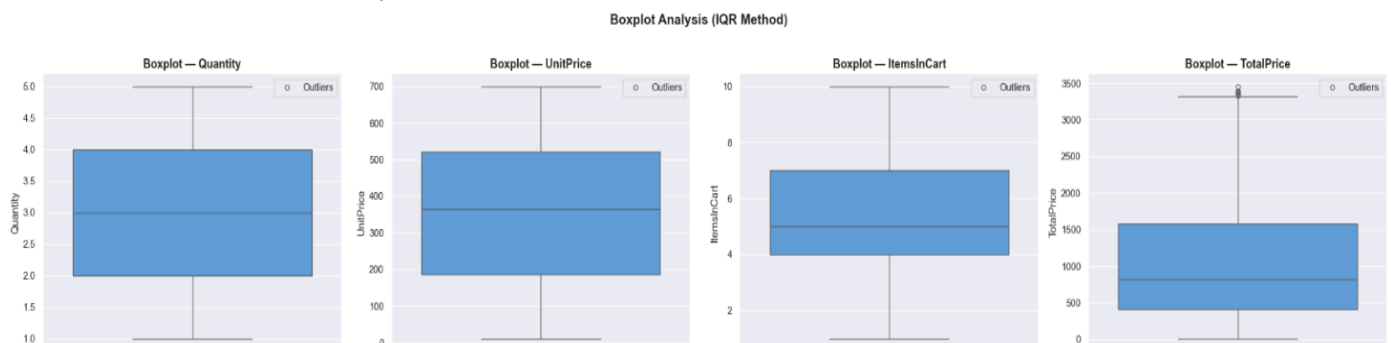
Histograms with KDE

Kernel Density Estimate (KDE) histograms were produced for each numeric column. Red dashed lines mark the mean and green dashed lines mark the median, making it straightforward to see the direction and degree of skew at a glance.



Boxplots (IQR Method)

Boxplots display the interquartile range (IQR), median and outlier positions for each numeric variable. Outlier points beyond the whiskers ($Q1 - 1.5 \times IQR$ and $Q3 + 1.5 \times IQR$) are rendered in red for visual clarity.



IQR Outlier Detection – Summary

For each numeric column the following were computed programmatically:

```
IQR OUTLIER DETECTION
Quantity
Q1=2.00 Q3=4.00 IQR=2.00
Fences: [-1.00, 7.00]
Outliers detected: 0

UnitPrice
Q1=186.06 Q3=521.57 IQR=335.51
Fences: [-317.20, 1024.83]
Outliers detected: 0

ItemsInCart
Q1=4.00 Q3=7.00 IQR=3.00
Fences: [-0.50, 11.50]
Outliers detected: 0

TotalPrice
Q1=410.52 Q3=1578.47 IQR=1167.95
Fences: [-1341.41, 3330.41]
Outliers detected: 8
```

Trend & Pattern Analysis

The Date column was parsed to datetime format, and the following time features were extracted for trend analysis: Month (numeric), Month Name (abbreviated string), Year and Day of Week.

Monthly Revenue Trend

Revenue was aggregated by year-month period to construct a line chart showing the revenue trajectory over the full date range. An area fill was added beneath the line to emphasize the magnitude of revenue at each period. This chart is the primary tool for identifying seasonal peaks, growth phases and revenue dips.

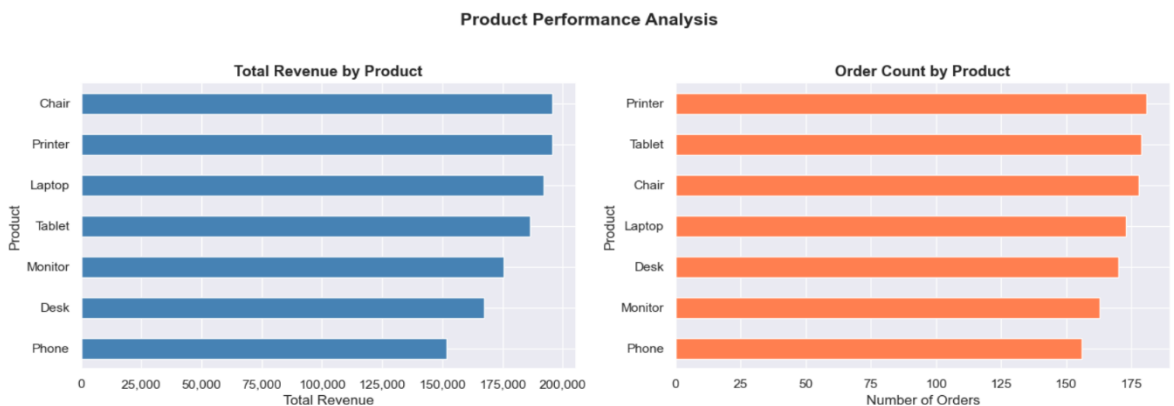


Product Performance

Two side-by-side horizontal bar charts were produced:

- Total Revenue by Product - identifies the highest-grossing products
- Order Count by Product - identifies the most frequently ordered products

Comparing these two charts reveals whether high-revenue products are also high-volume (indicating consistent demand) or whether a small number of large orders drive their revenue figure.



Referral Source Performance

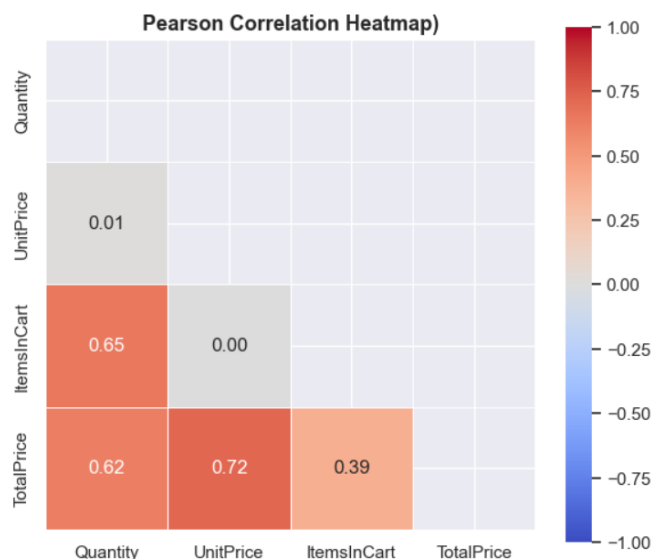
Performance was broken down by acquisition channel (ReferralSource) across three KPIs:

Referral Source Performance:			
ReferralSource	Total Revenue	Order Count	Avg Order Value
Instagram	275285.45	259	1062.88
Email	261808.55	250	1047.23
Google	250441.48	241	1039.18
Facebook	250410.90	228	1098.29
Referral	226815.58	222	1021.69

Correlation Analysis

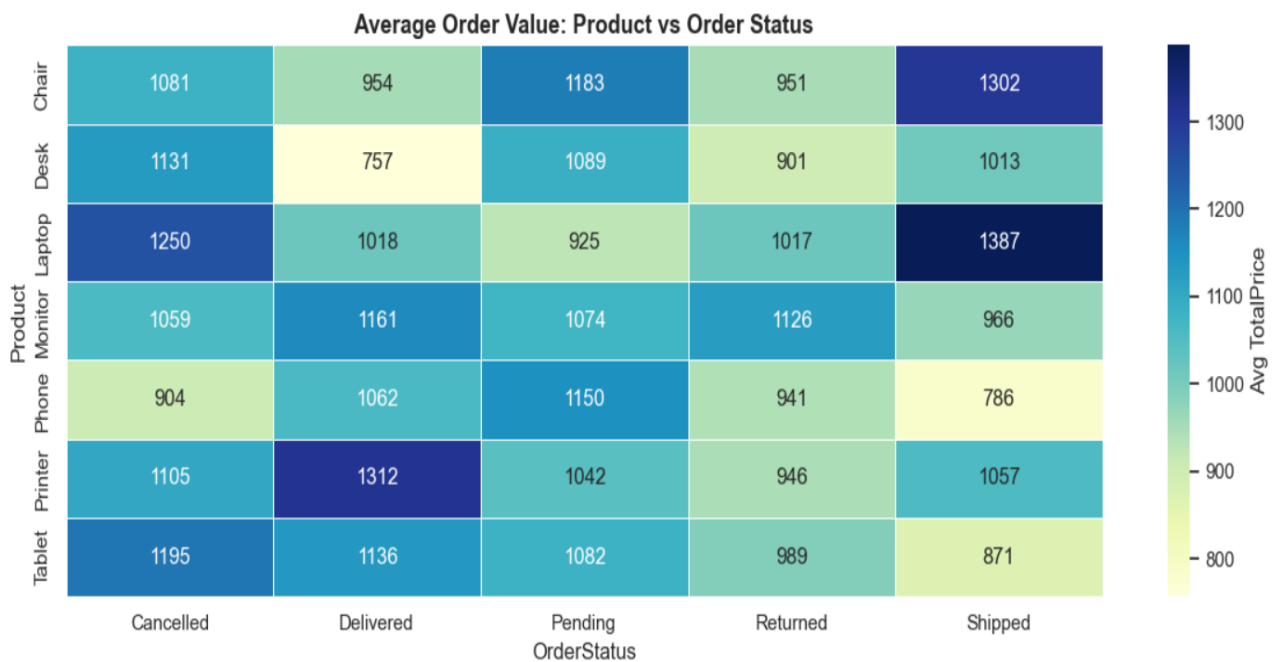
Pearson Correlation Heatmap

A Pearson correlation matrix was computed across all numeric columns and displayed as a lower-triangle heatmap (to avoid redundancy). The coolwarm diverging palette maps values from -1 (strong negative correlation) through 0 (no linear relationship) to +1 (strong positive correlation). Annotated cell values enable precise reading of each coefficient.



Product vs Order Status Pivot Heatmap

A pivot table was constructed showing the average TotalPrice for each combination of Product and OrderStatus. The resulting heatmap (YlGnBu palette) reveals which product-status combinations generate the highest average revenue; for example, whether completed orders for a specific product significantly outperform cancelled orders in monetary terms.



Conclusion

This EDA notebook provides a systematic analytical foundation for the sales dataset. The key outcomes of the analysis are:

- Descriptive statistics identify the central tendency and spread of all numeric variables, with automatic skewness flagging to highlight non-normal distributions.
- Distribution plots reveal whether data is symmetric, right-skewed or left-skewed and quantify how many records fall outside normal bounds using the IQR method.
- Monthly revenue trends expose seasonality and growth patterns across the observation period.
- Product and referral source breakdowns identify the highest-value and highest-volume segments for strategic focus.
- Correlation analysis quantifies linear relationships between numeric variables, while the pivot heatmap adds a categorical dimension (Product vs OrderStatus) to revenue understanding.

These findings can directly inform business decisions around inventory management, marketing channel investment, product prioritization and customer acquisition strategy.